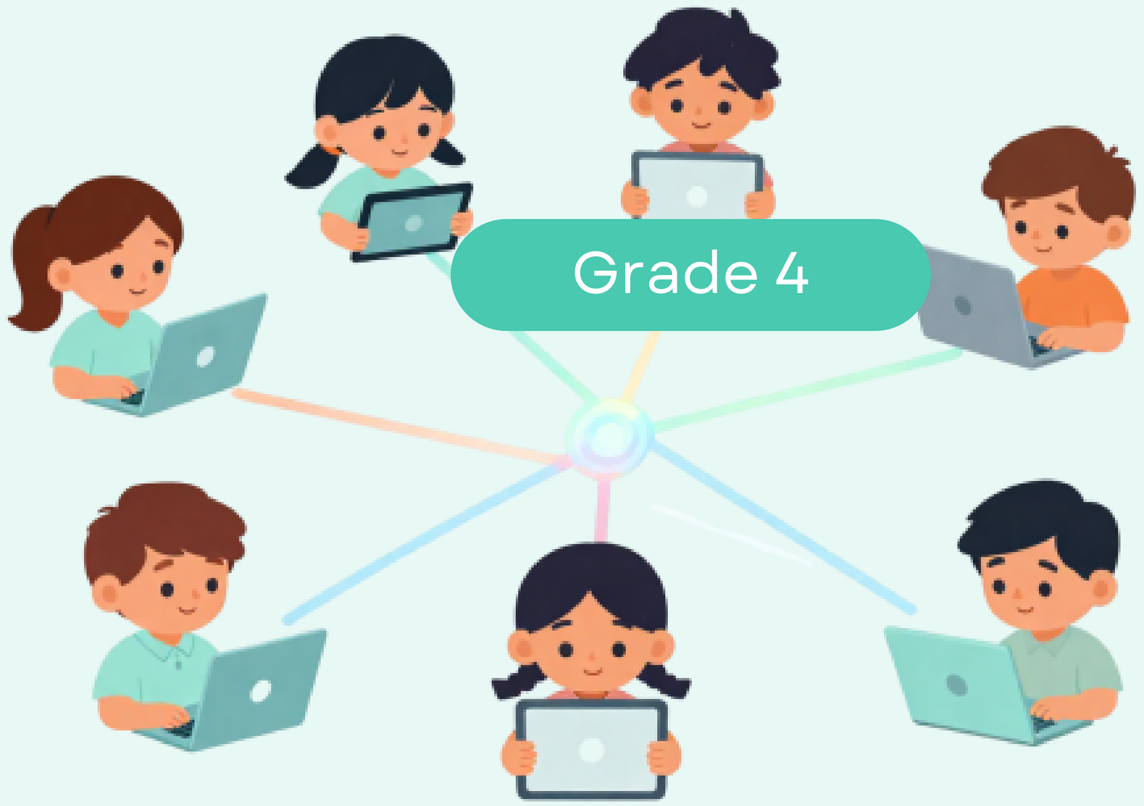
How the Internet Works

TCP – The Internet's Delivery System

Grade 4

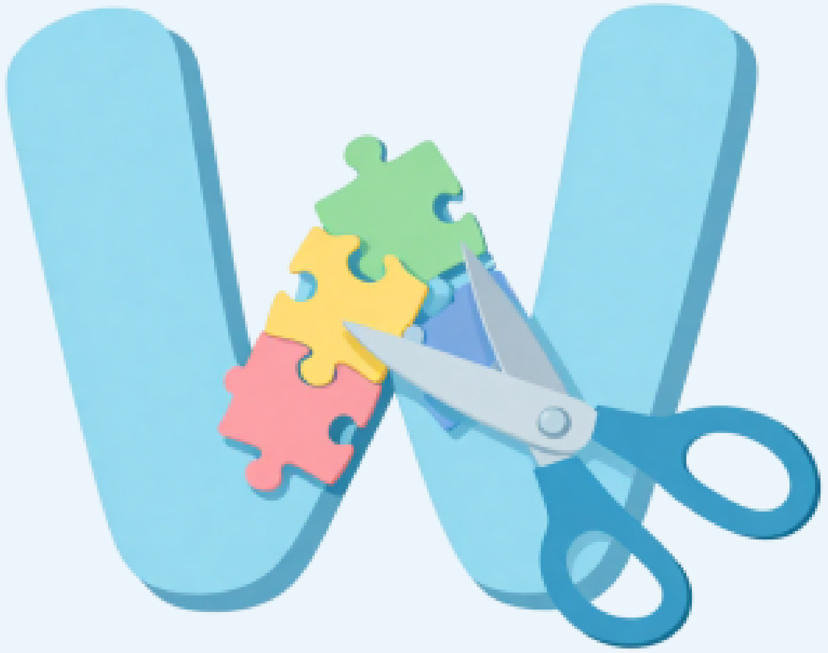


What is the Internet?

The internet is a giant network of computers all around the world, connected by cables, Wi-Fi, and satellites. It lets us share messages, pictures, videos, and more!

Sending a Message

When you send a message online, it doesn't travel in one piece. It gets broken into tiny parts called PACKETS – like cutting a letter into puzzle pieces!



What is TCP?

TCP stands for Transmission Control Protocol.

It's like a set of rules that computers follow to send and receive data correctly and completely.

TCP's Job

TCP makes sure:

1. Data is broken into packets
2. Each packet is numbered
3. All packets arrive safely
4. Packets are put back in order

Step 1: Break It Up!

Your message or file is split into small packets. Each packet gets a number so the receiving computer knows the correct order.

Step 2: Address Each Packet

Each packet gets a label – like writing an address on an envelope. The label says:

- Where it came from (your computer)
- Where it's going (the other computer)
- What number it is (so they go in order!)

Step 3: Send the Packets

The packets zoom through the internet – but they don't all take the same path!

Some go through cables under the ocean, some bounce off satellites, some travel through Wi-Fi. They race to the destination by the fastest route available!

Step 4: Check & Confirm

When packets arrive, TCP checks every single one:

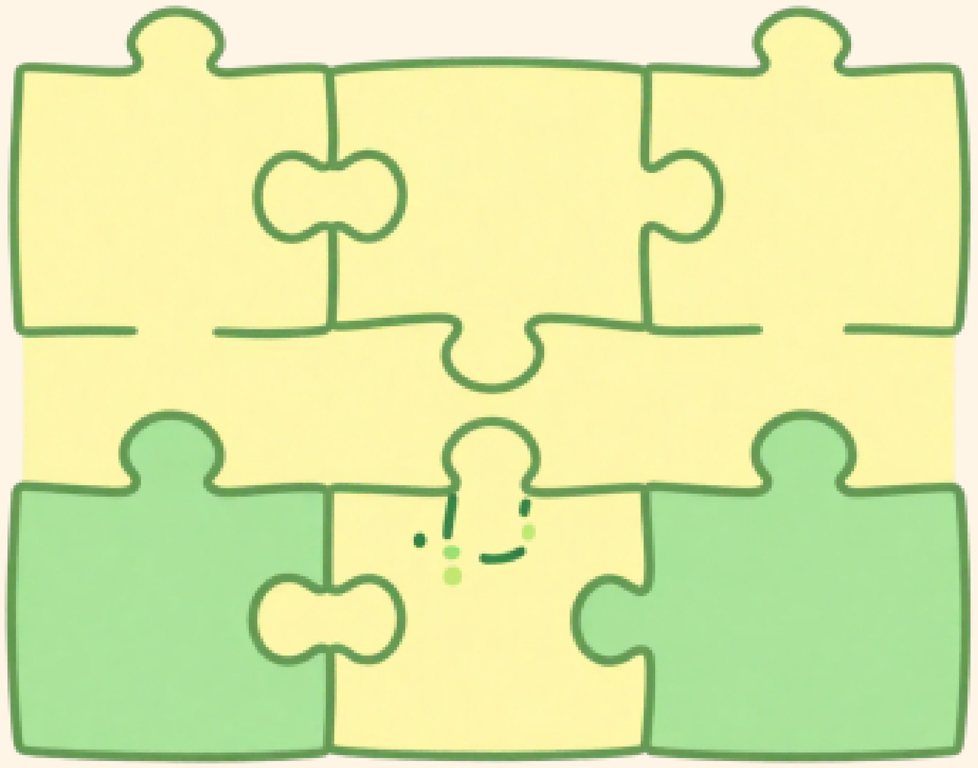
✓ Did all packets arrive?
✓ Are they in the right order?
✓ Is any packet damaged?

If something is missing, TCP says: "Please send that again!"

Step 5: Put It Together

Once all packets arrive safely, TCP puts them back together in the correct order – like solving a jigsaw puzzle!

Now your friend sees the full message, picture, or video exactly as you sent it. 🎉



The Three-Way Handshake

Before sending data, TCP does a "handshake" to make sure both computers are ready:

1️⃣ Computer A says: "Hey, can we talk?" (SYN)
2️⃣ Computer B says: "Sure, I'm ready!" (SYN-ACK)
3️⃣ Computer A says: "Great, let's go!" (ACK)

Now they can start sending packets!

TCP vs Sending a Letter

Sending a letter:

1. Write the letter
2. Put it in an envelope with an address
3. Post office delivers it
4. Receiver opens it

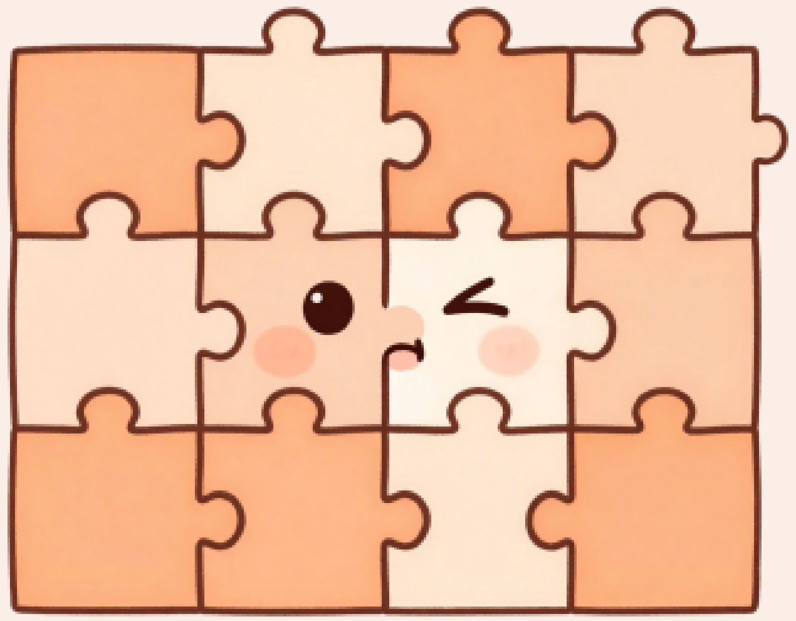
TCP does the same with data packets!

What Happens Without TCP?

Without TCP, your data could:

• Arrive in the WRONG order (jumbled puzzle!)
• Get LOST along the way
• Arrive with ERRORS nobody notices

TCP makes sure none of that happens!



TCP in Real Life

TCP is used every time you:

- Send a WhatsApp message
- Watch a YouTube video
- Load a website
- Send an email

It works silently in the background!

Fun Fact!

TCP was invented in 1974 by Vint Cerf and Bob Kahn.

They are called the 'Fathers of the Internet'!

Every single day, TCP delivers BILLIONS of packets around the world without you even noticing.

What Did We Learn?

Today we learned:

- Data travels as small PACKETS
- TCP breaks, labels, sends & checks packets
- The Three-Way Handshake starts every connection
- TCP makes sure nothing is lost or jumbled

You now know how the internet delivers your messages!